**Answers**

Exit quiz

# Solving simple linear equations with an additive step

**1** What is the value of the unknown in this bar model? $x =$ __10__ Fill in the blank

| $x$ | 3 |
|---|---|
| 13 | |

**2** Which best describes the mistake that has been made when solving the equation below? Tick **1** correct answer

$$\begin{aligned} x - 8 &= 15 \\ x - 8 + (-8) &= 15 + (-8) \\ x &= 7 \end{aligned}$$

- ☐ 15 add (−8) does not equal 7
- ☐ The same operation has not been applied to both sides of the equation.
- ☑ The correct additive inverse has not been added.
- ☐ The (−8) has not been added to the $x$ term and the constant term on the LHS.

**3** The solution to the equation $x - 5 = 15$ is when $x$ = __20__ Fill in the blank

**4** Which of these shows the correct working to solve the equation $x + 3.8 = 2.3$ in one additive step? Tick **1** correct answer

☐
$$\begin{aligned} x + 3.8 &= 2.3 \\ x + 3.8 + (-3.8) &= 2.3 \\ x &= 2.3 \end{aligned}$$

☐
$$\begin{aligned} x + 3.8 &= 2.3 \\ x + 3.8 + (-3.8) &= 2.3 + (-2.3) \\ x &= 0 \end{aligned}$$

☑
$$\begin{aligned} x + 3.8 &= 2.3 \\ x + 3.8 + (-3.8) &= 2.3 + (-3.8) \\ x &= -1.5 \end{aligned}$$

☐
$$\begin{aligned} x + 3.8 &= 2.3 \\ x + 3.8 + (-2.3) &= 2.3 + (-2.3) \\ x - 1.5 &= 0 \end{aligned}$$

**5** The solution to the equation $x + 7.2 = 14.6$ is when $x$ = 7.4 Fill in the blank

**6** Which of these is the solution to the equation $x - \frac{2}{3} = \frac{5}{6}$? Tick **1** correct answer

- [ ] $x = \frac{1}{6}$
- [ ] $x = \frac{3}{4}$
- [ ] $x = \frac{7}{9}$
- [ ] $x = 1$
- [x] $x = \frac{3}{2}$